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Quantifying Global Material Displacement from Surface Mining

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Corresponding Author:	Kamila Svobodova The University of Queensland Centre for Social Responsibility in Mining AUSTRALIA
First Author:	Vítězslav Moudrý
Order of Authors:	Vítězslav Moudrý
	Kateřina Gdulová
	Victor Maus
	Anna F. Cord
	Marie Lachaise
	Petra Pracná
	Alexey V. Alekseenko
	Kamila Svobodova
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Authors: Vítězslav Moudrý¹, Kateřina Gdulová¹, Victor Maus^{2,3}, Anna F. Cord⁴, Marie Lachaise⁵, Petra Pracná¹, Alexey V. Alekseenko⁶, Kamila Svobodova⁷

¹Faculty of Environmental Sciences, Czech University of Life Sciences Prague, Kamýcká 129, Prague-Suchdol, 165 00, Czechia

² Institute for Ecological Economics, Vienna University of Economics and Business, Welthandelsplatz 1/D5, 1020 Vienna, Austria

³ Advancing Systems Analysis Program, International Institute for Applied Systems Analysis (IIASA), Schlossplatz 1, A-2361 Laxenburg, Austria

⁴ Agro-Ecological Modeling Group, Institute of Crop Science and Resource Conservation, University of Bonn, 53113 Bonn, Germany

⁵ Remote Sensing Technology Institute, German Aerospace Center (DLR), Oberpfaffenhofen, Münchener Str. 20, 82234 Weßling, Germany

⁶ Institute for Integrated Management of Material Fluxes and of Resources (UNU-FLORES), United Nations University, Ammonstrasse 74, 01067 Dresden, Germany

⁷ Centre for Social Responsibility in Mining, Sustainable Minerals Institute, The University of Queensland, Saint Lucia, QLD 4072, Australia.

Corresponding authors:

Vítězslav Moudrý (moudry@fzp.czu.cz); Faculty of Environmental Sciences, Czech University of Life Sciences Prague, Kamýcká 129, Prague-Suchdol, 165 00, Czechia

Kamila Svobodova (k.svobodova@uq.edu.au); Centre for Social Responsibility in Mining, Sustainable Minerals Institute, The University of Queensland, Saint Lucia, QLD 4072, Australia.

Abstract

We quantify continental and national contributions to material displacement between 2011–2015 and 2016–2022. Globally, 279 billion m³ of material were extracted and 236 billion m³ deposited. Extracted material volumes are dominated by Asia (133 billion m³), followed by South America (62), North America (30), Australia (21), Africa (17), Europe (15), and Oceania (1). The highest extraction volumes occur in Indonesia, China, Russia, Australia, and Brazil. However, when normalized by country area, mining intensity is higher in smaller countries such as Guyana, Suriname, Ghana, and Germany. Mining is highly concentrated in specific biomes, including African montane grasslands and mangroves, South American deserts and xeric shrublands, Australian temperate forests, and Asian tropical forests, where extraction volumes exceed expectations based on geographic extent by a factor of two to six. Extracted and deposited volumes provide a complementary measure of mining intensity to site area, enabling consistent cross-site comparisons for global monitoring.

Keywords: Change Map, DEM of Difference, Global Dataset, Mining Footprint, Mining Intensity, Remote Sensing, TanDEM-X

1. Introduction

The global demand for raw materials has grown rapidly in recent decades, driven by industrialization, infrastructure development, and energy transition. This escalating demand has intensified mineral extraction, particularly through surface mining, which is now recognized as an important driver of land use and geomorphological change globally (Cooper et al. 2018; Giljum et al. 2025). The expansion of existing mines and the opening of new operations have resulted in substantial shifts in land volumes and amplified both environmental and social pressures (Berthet et al. 2024; Brereton et al. 2024; Luckeneder et al. 2021).

This material displacement, encompassing both material extraction in open pits and the deposition of waste material on dumps, profoundly alters landforms and topography (Hendrychová and Kabrna 2016; Mulková and Popelková 2024),

contributes to deforestation and habitat loss (Giljum et al. 2022), disrupts hydrological systems (Northey et al. 2016), and is frequently associated with long-term contamination of soil and water resources (Alekseenko et al. 2025; Berka et al. 2025; Zerizghi et al. 2022). At the same time, it is a major driver of population displacement (Owen et al. 2024, 2021), cultural erasure (Svobodova 2025) and the disruption of community cohesion and people's place attachments (Svobodova et al. 2021).

The growing availability of satellite-based data has enabled the creation of global-scale datasets capturing the spatial extent of mining land use (Maus et al. 2020, Werner et al. 2020; Liang et al. 2021; Maus et al. 2022; Tang and Werner 2023). While these datasets advanced research on land-use patterns in mining regions (e.g. Yang et al. 2026), they provide no insight into the volumetric scale of mining activities. A more explicit understanding of mining volumes is necessary to reveal the complexity of the environmental and social degradation and the magnitude of potential long-lasting legacies (Nassar et al. 2022).

Existing research on mining volumes remains fragmented, site-specific and often non-comparable across studies. Daham et al. (2024), for instance, used remote sensing imagery to detect mining activities and quantify sand extraction volumes across the Ganges-Brahmaputra-Meghna River system. Yoshida et al. (2019) proposed a methodology combining digital elevation model and land cover maps to detect and estimate the volume of mining and landfilling in Germany. However, these approaches are not readily transferable which limits their use for global-scale estimates of mining volumes.

Advances in Earth observation, particularly the availability of global digital elevation models (DEMs) for different time periods, now provide unprecedented opportunities to monitor the volumetric dimension of surface mining. Comparing DEMs from different time periods, so-called DEMs of Difference (DoD) can capture surface lowering and deposition, including changes caused by mining (e.g. Williams 2012; Pacina et al. 2012; Pandey and Kumar 2014; Ross et al. 2016; Louw & Avtar 2025). This approach makes it possible to go beyond mapping the horizontal footprint of mining, as done by Maus et al. (2022) and Tang and Werner (2023), and to quantify the three-dimensional transformations of landscapes. By doing so, it opens the door to global assessments of extracted and deposited volumes.

Our study estimates the volumes of extracted and deposited materials resulting from surface mining activities worldwide by comparing terrain changes between two time periods (2011–2015 and 2016–2022). Specifically, we (1) quantify the volume of extracted and deposited material across continents, biomes, and countries, and examine these volumes relative to their respective areas; (2) demonstrate the added value of volume-based analysis for global environmental assessment by comparing volume-based rankings to those based on mining extent; and (3) characterize individual mining sites in terms of extracted and deposited volumes and assess the relationship between polygon area and material volume. By providing global-scale volume estimates, our study offers a first-of-its-kind indicator for mining intensity that delivers new insights into the 3D footprint of resource extraction. These estimates can serve as a valuable basis for improving future assessments of environmental and social impacts associated with surface mining.

2. Materials and methods

To analyze the volumes of extracted and deposited materials resulting from surface mining, we used two complementary global products: the global-scale mining polygons dataset ([Maus et al. 2022](#); version 2), representing mining areas, and the TanDEM-X 30m DEM Change Maps ([Lachaise and Schweissheim 2024](#)), which provide global elevation change information. To derive volume for each mining polygon, we spatially intersected the mining polygons with elevation change and calculated the net volume of extracted and deposited material. We assume that elevation changes detected within mining polygons predominantly reflect material extraction or deposition associated with surface mining activities (but see Data limitations and uncertainty in section 2.4).

2.1. Mining areas

The Global-scale mining polygons dataset ([Maus et al. 2022](#); version 2) consists of 44,929 polygons covering 101,583 km² of mining areas. Mining areas were identified using coordinates from the S&P Global Market Intelligence database and manually delineated through visual inspection of several satellite-based, cloud-free images

(Google Satellite Imagery, Microsoft Bing Imagery, and Sentinel-2). The polygons cover various land uses related to mining activities such as open cuts, waste dumps, and processing facilities but they do not distinguish between them (Maus et al. 2020; 2022). While this dataset represents the most comprehensive compilation of polygon data on mining sites to date, it still does not capture all existing mines worldwide (Tang and Werner 2023; Maus and Werner 2024). In Maus et al. (2022), the authors reported omission and commission errors of mines were 21% and 3%, respectively.

2.2. TanDEM-X 30m DEM Change Maps

The TanDEM-X 30m DEM Change Maps are derived from data collected by the TanDEM-X satellite mission, a joint initiative of the German Aerospace Center (DLR) and Airbus Defence and Space aimed at mapping the Earth's surface (Hajnsek et al. 2025). The mission consists of a constellation of two satellites flying in close orbit formation and acquiring radar images of the Earth's surface. In accordance with the mission's primary goal, the acquired data served as the basis for the development of several global digital elevation models (Zink et al 2014; Rizzoli et al. 2017), including the TanDEM-X 30m Edited DEM, which was produced using data acquired over a four-year period between December 2010 and January 2015 (Gonzalez et al. 2024). Since the satellites were still in good condition following the completion of the first DEM, additional global coverage was acquired between 2016 and 2022 to generate a new, timely independent, global DEM (Bachmann et al. 2021).

The TanDEM-X 30m DEM Change Maps at a 1 arc-second spatial resolution represent the most recent addition to the mission products. They are available in two versions: one showing the first DEM change and another showing the latest DEM change with respect to the TanDEM-X 30m Edited DEM (Lachaise et al. 2022; Lachaise and Schweissheim 2024). In this study, we used the last DEM Change Map, which represents elevation changes between the TanDEM-X 30m Edited DEM (acquired between 2010 – 2015), and most recent data acquired between 2016 and 2022 (hereafter referred to as DCM).

2.3. Global mining volume assessment per continent, country and biome

We assessed the volume of extracted material and the volume of deposited material within all mining polygons. To differentiate between extracted and deposited volumes, we treated cells with positive and negative values in the DCM separately. To calculate

extracted and deposited volume, we summed the elevation differences of all cells in the DCM within each mining polygon and multiplied the result by the area of a single cell. Volume estimates therefore represent change between 2010–2015 and 2016–2022, aggregated at the spatial resolution of the TanDEM-X 30 m DEM Change Maps and constrained by mining polygon boundaries.

We quantified the volumes of both extracted and deposited materials, at the levels of continents, countries, and individual mining sites. In addition, the extracted and deposited values were normalized by total land area at continental and country levels, and by the total extent of mining areas at the continental level, to enable meaningful comparison among regions that differ in area and the extent of mining activities. Country and continent total areas used for normalization were obtained from World Bank Open Data (<https://data.worldbank.org/>). To highlight differences between areal and volume metrics, we compared rankings based on extracted volumes with those based on the total area of mining polygons. Furthermore, we analyzed patterns across the 14 terrestrial biomes of the world (regions with floristic similarity, similar climatic factors and vegetation structure), seven of which are forested and seven non-forested, using the biome map by [Dinerstein et al. \(2017\)](#).

For each mining polygon, we compared extracted and deposited volumes and summarized for each continent the number of polygons where extraction exceeded deposition, deposition exceeded extraction, both were zero, or only one of the two occurred. We also evaluated the relationship between mining polygon area and material volume for both deposited and extracted materials. Finally, we derived a global grid representing extracted and deposited volumes at a spatial resolution of 30 arcminutes (approximately 55 km at the equator), calculated as the total volume of material extracted or deposited within each grid cell, respectively.

2.4. Data limitations and uncertainty

Volume estimates were derived from elevation differences, and their accuracy depended on land cover, acquisition timing, and DEM editing procedures. To provide additional context for interpreting the estimated volumes and to account for potential sources of uncertainty, we used three auxiliary layers provided with the TanDEM-X 30m Edited DEM and TanDEM-X 30m DEM Change Maps, respectively. First, we used a land cover mask (LCM) to assess areas of forest, water, and land within

individual polygons. Knowledge of forest presence is especially important because, given the nature of the data, forest height may be included in estimates of extracted volume. In addition, temporal information is important because acquisition dates affect the time span over which volume changes are estimated. Therefore, we used the acquisition date (DATE) for each cell of the second global coverage acquired between 2016 to 2022. Finally, we adopted the editing mask (EDM) to indicate in which polygons alternative DEMs were used, which affects the time period for which the volume of extracted and deposited material was estimated. All auxiliary layers were summarized at the mining polygon level, providing contextual information to assess the reliability and potential uncertainty of volume estimates for each individual polygon.

3. Results

3.1. Extracted and deposited material volumes across continents

Worldwide, the total volume of extracted material based on the DEM Change data was 279,121 million m³ while the volume of deposited material was 236,701 million m³. Considerable differences between the continents can be observed (Figures 1 and 2). Asia dominates in terms of both extracted and deposited material, with extraction volumes 2.1 times higher than South America, which ranks second, and 8.6 times more material than Europe. Deposited volumes in Asia are 2.7 times higher than in South America and 7.1 times higher than in Europe. South America and North America show similar deposited volumes; however, South America extracted more than twice the material extracted in North America. In North America and Australia deposited volumes exceeded extracted ones, in contrast to Asia, and South America, where extraction was higher (Figure 1a). A comparison of the volumes of extracted and deposited material relative to each continent's total area showed the most intensive mining in South America, followed by Asia and Australia, while the lowest intensity was observed in Africa (Figure 1b). For example, in Europe, the total extracted volume of 15 billion m³ is equivalent to digging a hole 15x10x10 meters in size per every square kilometer of the continent. In contrast, Asia's total extracted volume of 133 billion m³ would result in a hole nearly twice as large. A comparison of the volumes of

extracted and deposited material relative to the area of mining sites showed that the depth of the mines (and the height of the mine waste heaps) increased on average by 2 to 4 meters per square meter of mining areas, depending on the continent. The greatest increase in depth (or height) was observed in Europe, while the smallest was identified in Africa (Figure 1c).

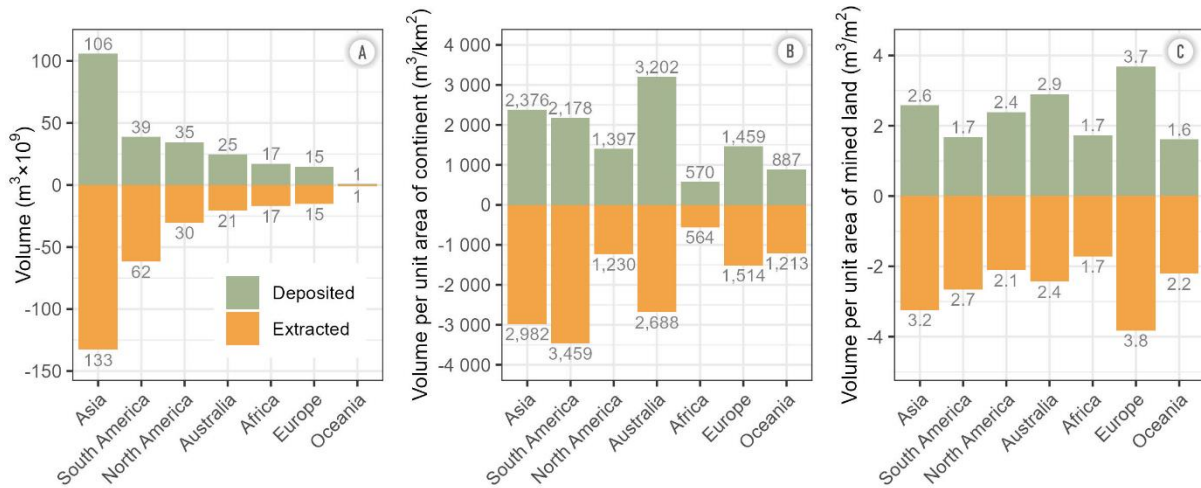


Figure 1. Total volumes of extracted and deposited material (billions of m^3) by continent (A), volumes of extracted and deposited material normalized by continent area (B), and volumes normalized by the total area of mined land on each continent (C). Volumes represent the period from 2010-2015 to 2016-2022.

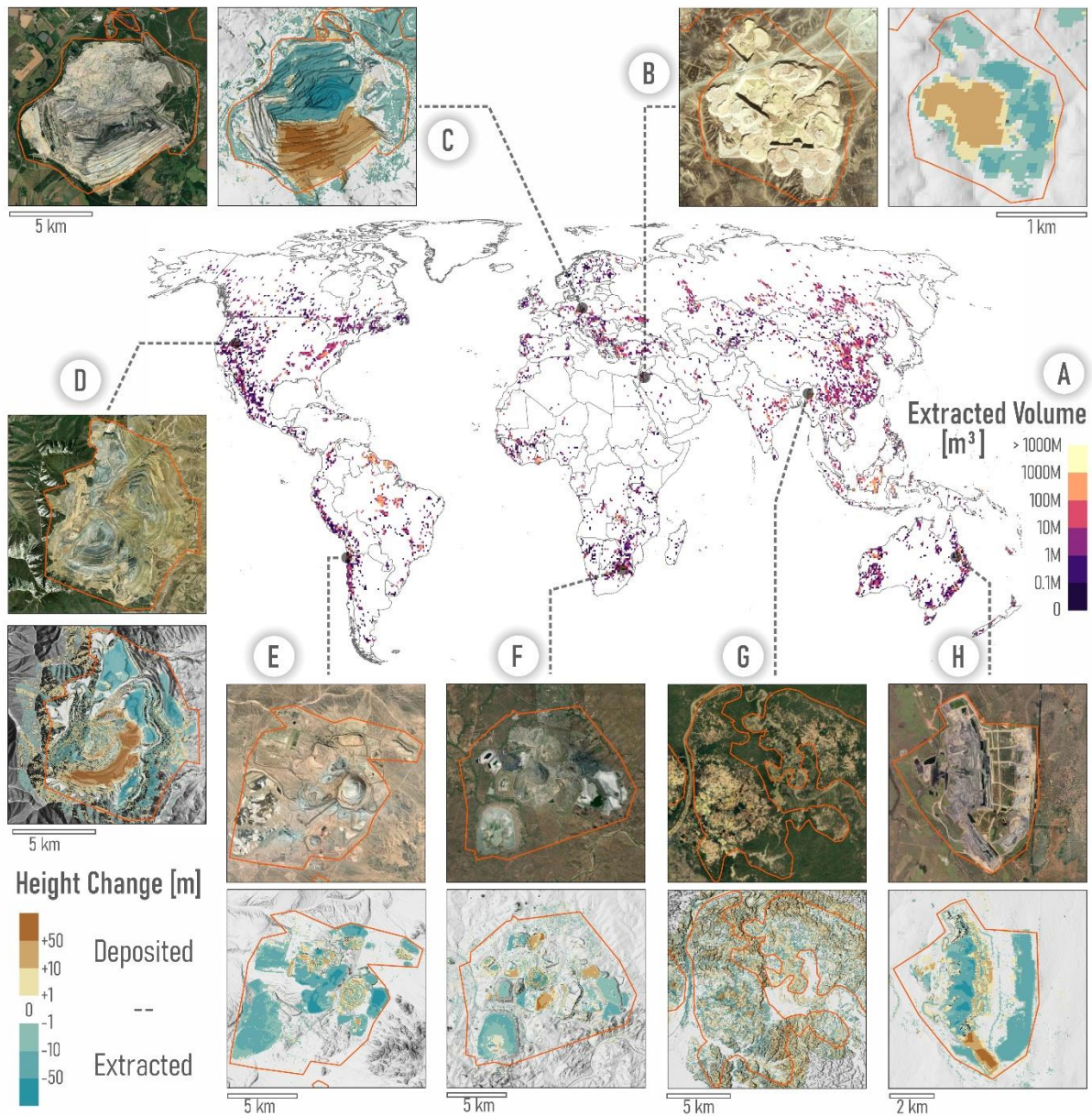


Figure 2. Global map of extracted material volume, expressed as the total volume extracted within each 30-arcminute grid cell (A). Images (B–H) show mining areas and corresponding TanDEM-X DEM Change Maps for seven representative sites across all continents: (B) Jordan (31.05°N, 36.12°E); (C) Poland (50.91°N, 14.89°E); (D) USA (40.53°N, 112.14°W); (E) Chile (22.98°S, 69.06°W); (F) South Africa (23.99°S, 31.13°E); (G) Myanmar (24.85°N, 95.32°E); (H) Australia (23.39°S, 148.96°E).

3.2. Biome-specific patterns of material extraction

Forested biomes exhibit more than twice as much the extraction volume (190 million m³) as non-forested biomes (89 million m³) (Figure 3); however, the spatial extent of mining areas between these two categories differs much less, covering 56 and 45 thousand km², respectively (Figure S1). Globally, the most affected biome is the Tropical & Subtropical Moist Broadleaf Forests, both in terms of total extracted volume (113 million m³) and total extracted area (29 thousand km²). The second most affected biome by volume is the Temperate Broadleaf & Mixed Forests (41 million m³), followed by the Deserts & Xeric Shrublands (26 million m³). However, in terms of area, their ranking is reversed, with areas of 11 thousand km² and 15 thousand km², respectively. Ranking by volume further emphasizes the disproportionate impact on Tropical & Subtropical Moist Broadleaf Forests: the extracted volume is 4.3 times higher than in the Deserts and Xeric Shrublands (Figure 3), whereas the area affected by mining is only 1.9 times larger (Figure S1).

In Asia, the highest extraction volumes are concentrated in Tropical & Subtropical Moist Broadleaf Forests, which account for 21% of the global total, followed by Temperate Broadleaf & Mixed Forests. Non-forested biomes, including grasslands and deserts, also contribute considerably. A similar pattern is observed in South America, where Tropical & Subtropical Moist Broadleaf Forests dominate (17% of the global extracted volume), with additional contributions from grasslands and deserts (Figure 3). In Europe and North America, extraction is likewise concentrated in forested biomes, particularly Temperate Broadleaf & Mixed Forests in Europe (6% of the global extracted volume) and Boreal Forests in North America (3% of the global extracted volume), although grasslands and deserts also represent important components of total extraction. In contrast, Australia is dominated by non-forested biomes, especially tropical grasslands and deserts. Africa shows a similar pattern, with grasslands as the primary biome affected, alongside notable contributions from tropical forests. Oceania exhibits comparatively low extraction overall, mainly confined to tropical forest biomes (Figure 3).

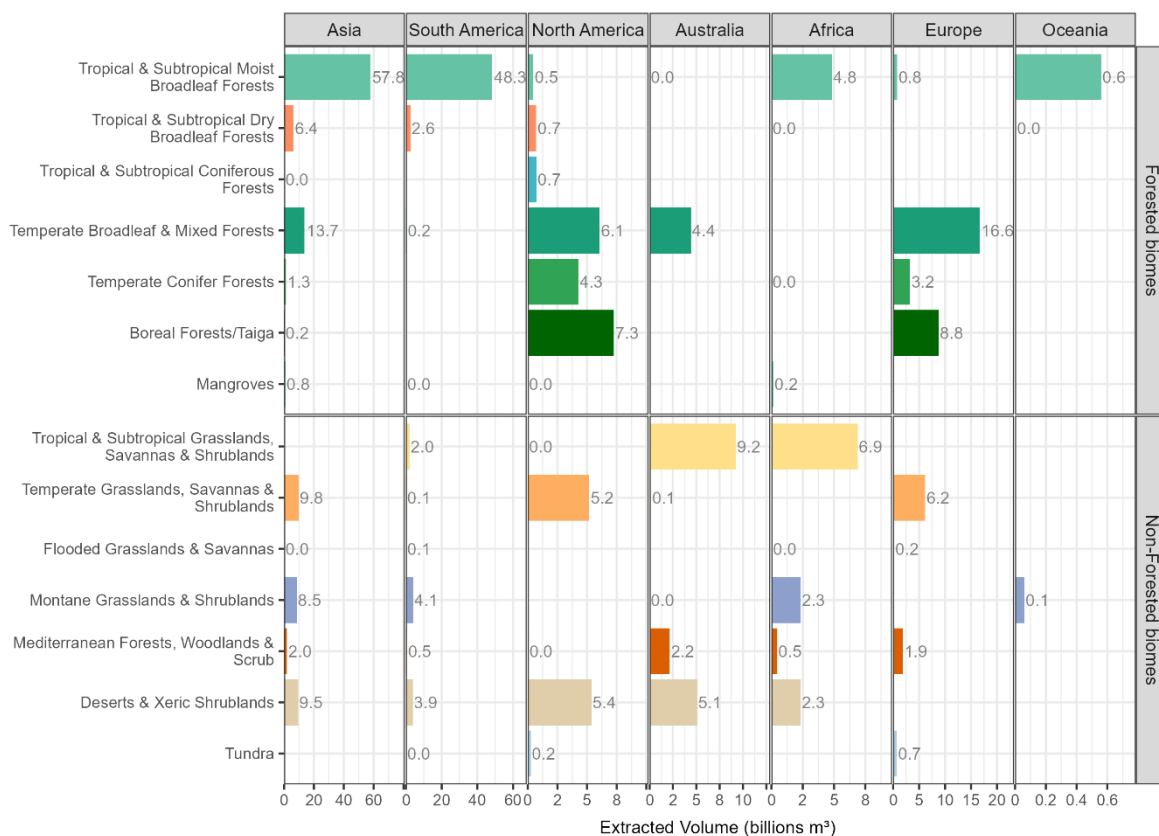


Figure 3. Mining extraction volumes across seven forested and seven non-forested biomes on each continent. Volumes represent the period from 2010-2015 to 2016-2022. Note the different scales on the x-axis. For comparison with mining area see supplementary material (Figure S1).

When mining volume is normalized by the share of a biome on each continent, clear hotspots emerge. Table 1 reveals that mining is highly concentrated in specific regions. The highest intensities occur in Africa's Montane Grasslands & Shrublands and Mangroves, as well as South America's Deserts & Xeric Shrublands, where extraction volumes are four to six times higher than expected based on geographic extent. High intensity is also observed in Australia's Temperate Broadleaf & Mixed Forests and Asia's Tropical Moist Broadleaf Forests, suggesting that these biomes serve as primary mineral hubs for their continents. In contrast, biomes such as the Tundra and Flooded Grasslands consistently show the lowest mining intensity worldwide (Table 1).

3.2. Extracted and deposited material volumes across countries

An analysis of the total volume of extracted material by country revealed that the majority (56%) was concentrated in just six countries: Indonesia, China, Russian Federation, Australia, Brazil, and United States. Another 14 countries together accounted for 32% of the total extracted volume, while the remaining 125 countries contributed only 12% (Figures 4a and 5). Similarly, 56% of the deposited material was accounted for by five countries as well; however, this group differed slightly, with Canada included instead of Brazil (Figure 4a and 5). When normalized by country area, however, the data revealed a higher intensity of mining activities in smaller countries such as Guyana, Suriname, Ghana, and Germany (Figures 5 and 6). Indonesia was the only country to be ranked in the top six for both overall volume (Figure 5) and volume normalized by country area (Figure 6). Note, that the ranking according to extracted volume differed from the ranking based on the area of mining sites (Figure 4b).

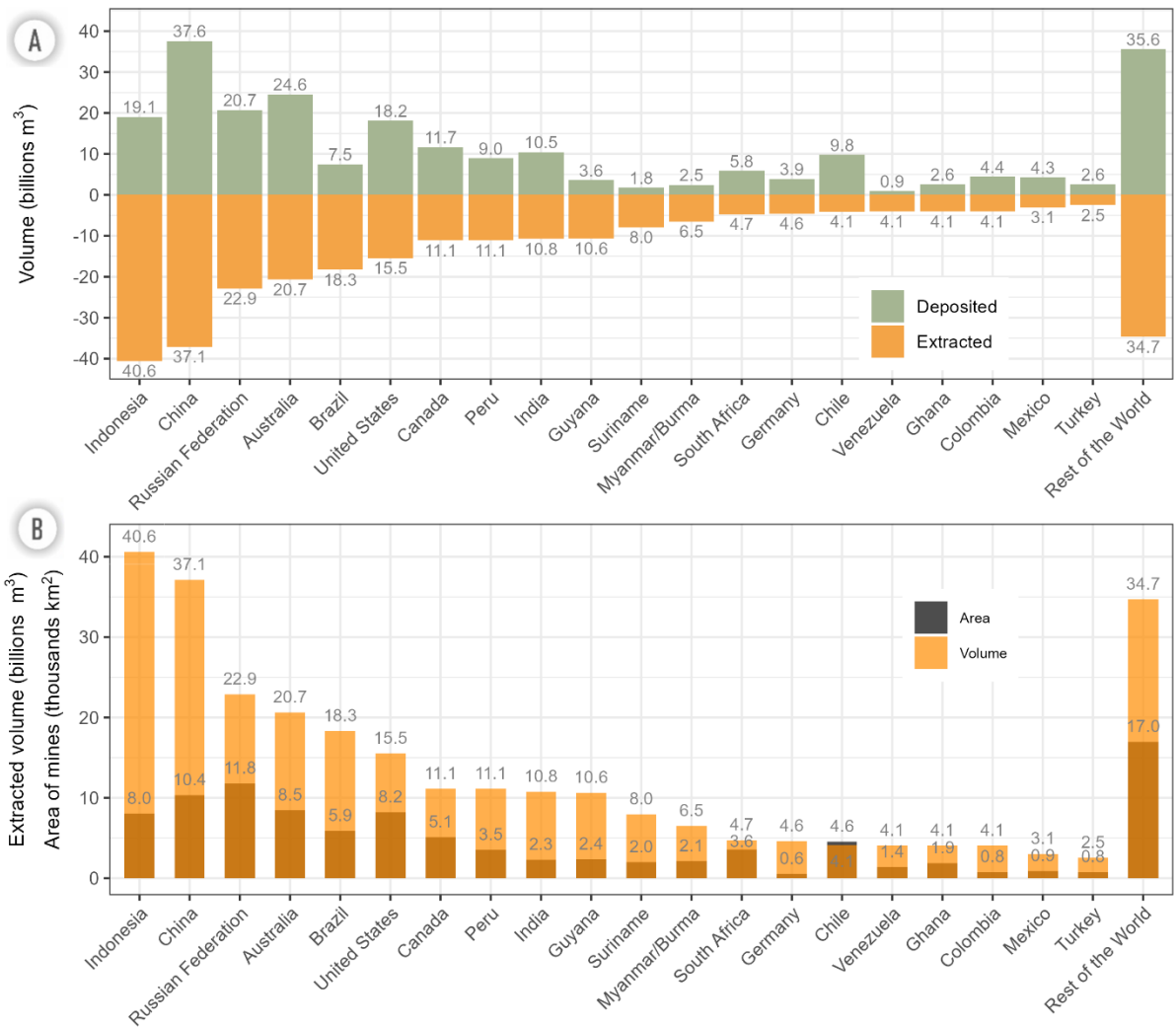


Figure 4. The volume of extracted and deposited material (billions of m³) by country (A). Extracted volume per country in millions of m³ compared to mining land area in square kilometers (B). Countries are sorted by the volume of extracted material. Volumes represent the period from 2010-2015 to 2016-2022.

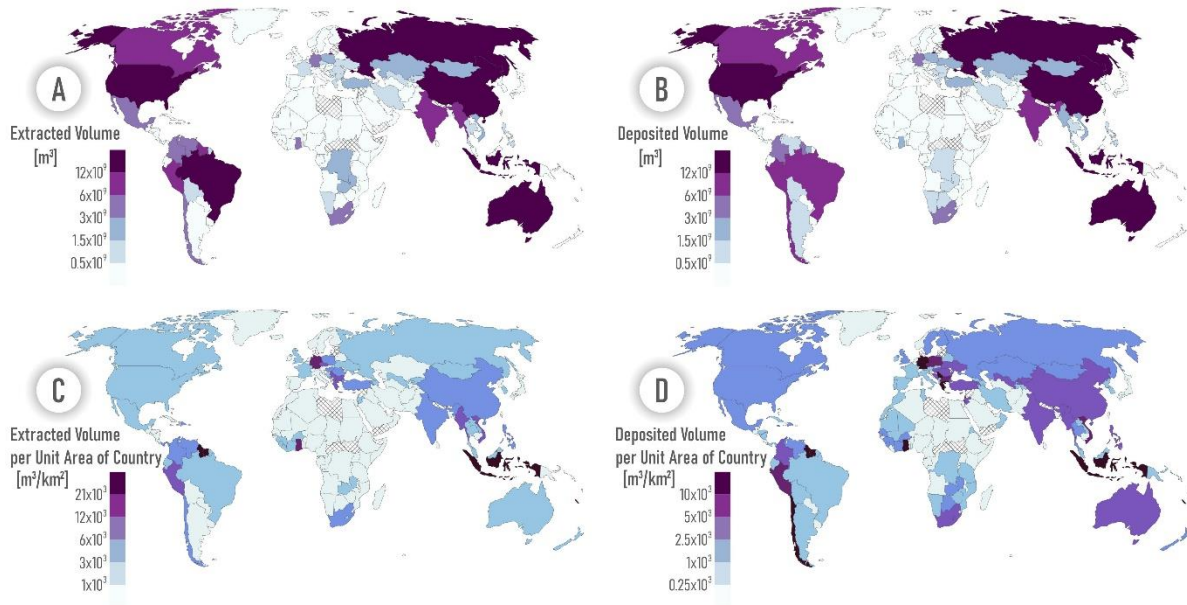


Figure 5. Global patterns of mining volumes. The volume of extracted (A) and deposited (B) material (billions of m^3) by country. The volume of extracted (C) and deposited (D) material relative to a country's area. Volumes represent the period from 2010-2015 to 2016-2022.

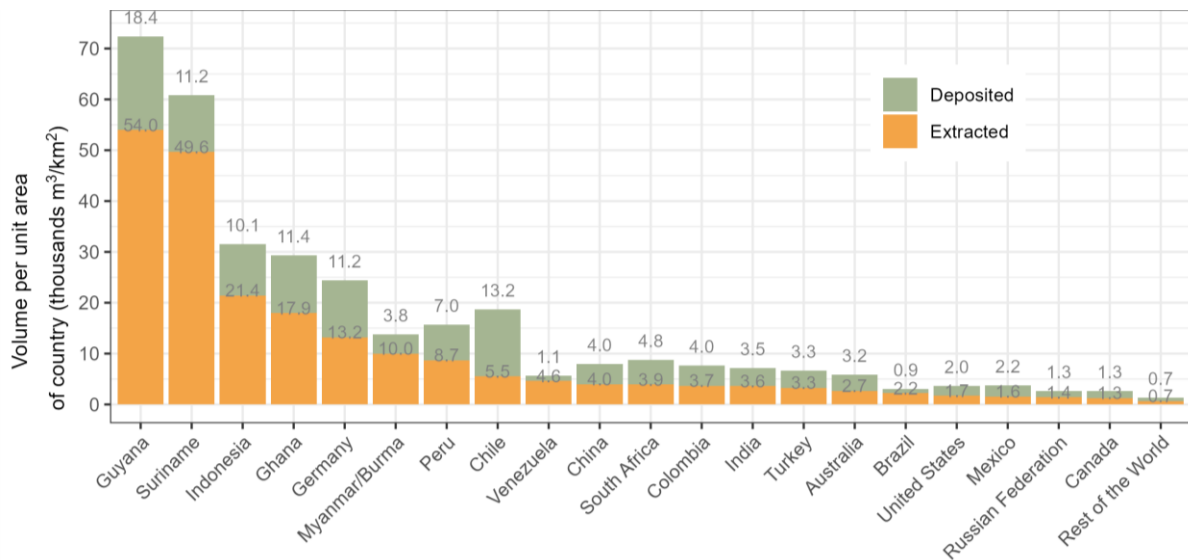


Figure 6. The volume of extracted and deposited material relative to a country's area. The countries are sorted by the extracted material. Volumes represent the period from 2010-2015 to 2016-2022.

3.3. Individual mining sites

Analysis of individual mining polygons showed that, of the 44,929 mining polygons, 87% (39,125 polygons) had non-zero values for both deposited and extracted material. A total of 2,270 polygons had zero deposited material, while 2,027 polygons had zero extracted material. Of the 39,125 polygons with non-zero values for both variables, 55% (21,693 polygons) had extracted volumes exceeding deposited, 44% (17,057 polygons) had deposited volumes exceeding extracted volumes, and 1% (375 polygons) had equal extracted and deposited volumes. Most polygons are small (<1 km²) and associated with relatively low volumes of extracted and deposited material (<1 million m³), whereas a small number of large polygons account for disproportionately high volumes (Figure 7).

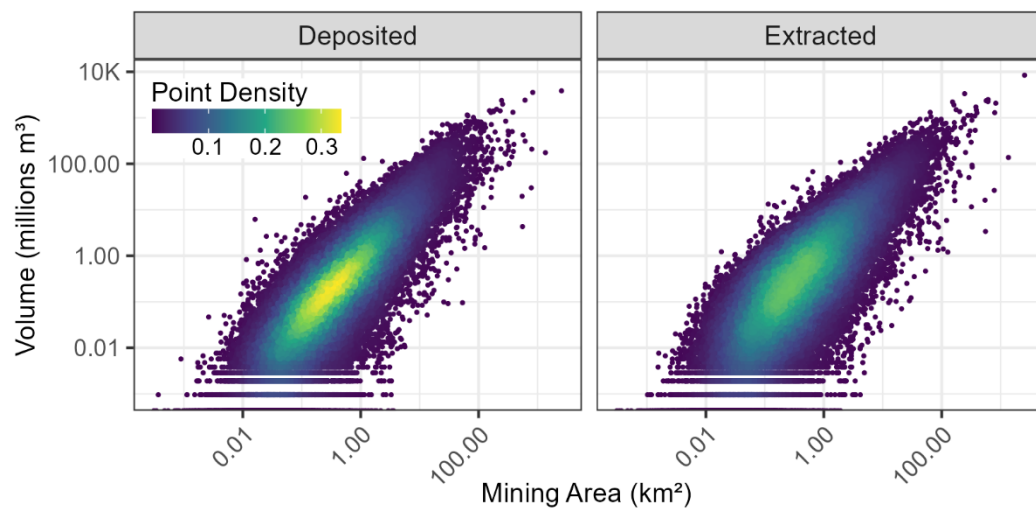


Figure 7. Relationship between mining polygon area and volumes of extracted and deposited materials. Volumes represent the period from 2010-2015 to 2016-2022.

4. Discussion

By analyzing global data on extracted and deposited mining volumes, we reveal that mining intensity is uneven across continents and concentrated in specific, often sensitive biomes and countries with weaker governance and lower human development. These disparities are associated with potential social and environmental risks.

Our study further demonstrates the potential of TanDEM-X 30 m DEM Change Maps to analyze global mining sites and their associated volumes. Extracted and deposited mining volumes provide a new quantitative indicator of mining intensity. Together with mining area, these metrics can capture differences in disturbance magnitude and offer a more comprehensive measure of the global mining footprint (Maus and Werner 2024).

4.1 Continental and country-level disparities in mining volumes and associated social and environmental risks

The findings highlight pronounced continental and country-level asymmetries in the scale and intensity of extracted and deposited material associated with mining (Figures 1 and 2). Asia's dominance in both extracted and deposited volumes reflects its central role in global mineral supply chains and rapid industrialisation over the past decade (Kalra et al. 2026), and points to a concentration of large, high-throughput mining operations associated with substantial landscape transformation. Europe's high volume-to-area ratio in both material extraction and deposition suggests relatively more vertically intensive mining compared to other regions, while Africa's lower depth-to-area ratio indicates more extensive but shallower operations.

Our country-level analysis underscores the highly uneven global distribution of mining activity (Svobodova et al. 2022; Hendrix and Noland 2014). Just six countries, Indonesia, China, Russia, Australia, Brazil and United States, account for more than half (56%) of the total global extracted material (Figures 4 and 5). When deposited volume is considered instead of extracted volume, Canada replaces Brazil among the top contributors accounting for 56% of the total. This is consistent with Maus et al. (2022) who reported that the same six leading countries accounted for 52% of the total mapped mining area. All seven countries (including both Brazil and Canada) rank among the global top eight nations in terms of total production value of minerals, metals and coal (ICMM 2025).

In 81 out of 145 countries, extracted volumes exceeded deposited volumes (Figures 4 and 5). However, in countries such as Chile, Australia, United States, Mexico and South Africa, the deposited volumes were considerably higher. This likely reflects underground extraction not captured by mapped polygons and the bulking effect,

whereby overburden expands when excavated and relocated (Lv et al. 2023). In addition, commodity-specific bulking, influenced by properties such as grain size, moisture and compaction, might further amplify deposited volumes. Because bulking may distort the relationship between extracted and deposited material, these metrics are recommended to be interpreted separately.

When normalized by total country area, the highest mining intensity is observed in countries including Guyana, Suriname, Germany, Indonesia, Ghana, Myanmar, Peru, and Chile (Figure 6). In these countries, large extracted or deposited volumes can translate into substantial land transformation and infrastructure expansion and gradually bring mining operations and human settlements into closer proximity. One implication of this “boundary effect” (Owen et al. 2024) is that social impacts, including mine-induced community displacement, are more likely as mining lifecycles advance. This occurs as mining footprints extend beyond their initial permitting and lease conditions, often accompanied by diminishing levels of public accountability and due process (Owen and Kemp 2019). Both the growth in mine depths and the increasing volume of mine waste, along with its mobilization into surrounding environments (e.g., Macklin et al. 2023; Owen et al. 2020), amplify environmental pressures while simultaneously creating new interfaces with communities that may not have been included in initial consultation or consent processes.

Understanding interactions between mining volumes, communities and the surrounding environment is critical. Without mapping the points of their intersection, researchers and policymakers may continue to treat mines and their footprints in isolation, overlooking broader environmental and social implications that are essential for informing effective restoration and sustainable post-mining land-use planning (Edwards et al. 2014; Frantál et al. 2024; Measham et al. 2024; Mimba et al. 2023). At the same time, insights into how different national governance systems manage consent, permitting and social and environmental safeguards are key to understanding the implications of these spatial patterns. Our results indicate that the highest mining intensity occurs in countries with weaker governance and lower human development such as Myanmar, Ghana, Guyana, Suriname and Indonesia. In these contexts, intense mining can strain institutional capacity and, without robust governance frameworks and meaningful community engagement, may undermine key aspects of quality of life. Yet, there is currently no comparable set of policy innovations

addressing the interaction between mining footprints and social systems or safeguarding affected communities and ecosystems. This aspect of mining governance must shift from the margins of mining and development debates to their center.

4.2 Mining volumes in specific, often sensitive biomes

Forested biomes experience more than twice the extraction volume of non-forested biomes, despite having only moderately larger mined areas (190 vs. 89 billion m³ across 56 vs. 45 thousand km²; Figures 3 and S1). Tropical and Subtropical Moist Broadleaf Forests are the most heavily impacted biome in terms of total extracted volume (Figure 3). A different pattern emerges in South America's Deserts & Xeric Shrublands and Africa's Montane Grasslands & Shrublands, which experience disproportionately high volumes of extraction and deposition relative to their spatial extent (Table 1). This pattern indicates that these biomes are subject to intensive industrial activity far beyond what their proportional land area would imply. Indeed, research into land-use patterns shows that arid and remote biomes often experience lower competition from other sectors, facilitating the expansion of capital-intensive mining operations ([Bridge 2004](#); [Sonter et al. 2018](#)), as the opportunity cost of land in these regions is significantly lower than in temperate or fertile biomes ([Auty 2001](#)).

To a lesser extent, disproportionately high extraction and deposition are also observed in forested biomes, including mangroves, across several continents (Table 1). This raises important considerations for biodiversity conservation and land-use governance, as many forested biomes coincide with global biodiversity hotspots ([Myers et al. 2000](#)) and regions of high ecosystem service provision and climate change mitigation ([Dar et al. 2020](#); [Grace 2004](#)). Continental leaders in mineral production, such as Australia (Temperate Broadleaf & Mixed Forests) and Asia (Tropical Moist Broadleaf Forests), demonstrate how specific biomes may become the mineral hubs for an entire continent's economic output. This pattern is consistent with studies documenting particularly intensive mining activities in tropical forest regions, such as Brazil and Indonesia, causing considerable deforestation (e.g., [Siqueira-Gay and Sánchez, 2021](#); [Stanimirova et al. 2024](#); [Giljum et al. 2025](#)). Even though planetary estimations are difficult, as satellite images do not directly reveal the

commodities linked to deforestation in a specific mining area ([Kramer et al. 2021](#)), gold and coal mines are reported to dominate mining-related deforestation globally ([Zhang et al. 2025](#)). In comparison, the consistently low intensity in biomes like the Tundra or Flooded Grasslands suggests that extreme environmental conditions, permafrost, and logistical constraints may act as significant barriers to large-scale extraction ([White-Nockleby and Odell 2025](#)).

Finally, our findings raise important questions regarding the global supply chain and the ecological trade-offs. The concentration of mining activity in high-intensity biomes likely leads to localized habitat fragmentation and biodiversity loss ([White-Nockleby and Odell 2025](#); [Sonter et al. 2018](#)). From a governance and conservation perspective, our findings underscore an urgent need to focus regulatory oversight on ecologically particularly valuable ecosystems that may carry disproportionate mining impacts.

4.3 Data limitations

Uncertainty in extracted and deposited volumes may stem from errors in both input datasets. The Global-scale mining polygons dataset (version 2) is an incomplete representation of mining areas worldwide (see [Maus et al. 2022](#)), which affects the estimates of the volumes. Likewise, TanDEM- X 30m DEM Change Maps introduce uncertainty due to the limitations in measurement technology, differences in acquisition periods and variations in the original land cover of mining areas. To assess these limitations, we used three auxiliary datasets (EDM, LCM, DATE) provided with the TanDEM- X data. Additionally, biome maps operate at a coarse spatial resolution, meaning that mines located in agricultural land, savanna mosaics or degraded landscapes may still fall within the boundaries of a forested biome classification. This can inflate the apparent association between mining volumes and forested landscapes, as shown by [Särkinen et al \(2011\)](#).

The main source of uncertainty relates to the timing of data collection, which varies across polygons. The estimated volumes correspond to a period of approximately 6–10 years. Most TanDEM-X 30m DEM data for the first global coverage were collected between 2011-2013, and for the second coverage between 2018–2020, with minimal differences between continents (Figure S2). Some data were collected outside these periods, so elevation changes may introduce small discrepancies in assessments at

the country or polygon level. To account for this uncertainty, each polygon is assigned an attribute indicating the acquisition date for the second acquisition period. This is, however, not possible for the first acquisition period, because the final DEM represents an average of repeated acquisitions (i.e., two acquisitions in low-relief terrain, and up to about ten acquisitions in more complex mountainous terrain; [Gdulová et al. 2020](#)). In some areas, terrain data were occasionally edited (e.g., to fill voids) using earlier satellite sources from different periods, but our analysis using EDM shows this affected less than 1% of mining polygons.

In addition, TanDEM-X DEM provides a digital surface model (DSM), rather than a digital terrain model (DTM), meaning that it includes vegetation height in addition to the ground surface ([Bielski et al. 2024](#)). This can inflate estimates of excavation volume, especially in forested areas, as the changes derived may reflect both logging and mining. For example, clearing a 1 km² forest with 40 m trees could appear as 30–35 million m³ of extracted material, depending on the depth of radar penetration into the canopy ([Gdulová et al. 2021](#); [Schlund et al. 2023](#)). To assess this issue, we used a land cover mask (LCM) for the first acquisition period (2010-2015) to identify forested areas within mining polygons. Some higher extraction volumes in tropical and subtropical forests (Figure 3) may reflect this vegetation bias (Figure S3). Nonetheless, even for originally forested polygons, our findings still capture the intensity of mining and the substantial deforestation caused over the assessed period (e.g., [Barenblitt et al. 2021](#); [Quash et al. 2024](#); [Giljum et al. 2025](#)).

4.4 Future research

Our results highlight both the potential and the limitations of using global DEMs to quantify mining impacts worldwide. Future work could incorporate earlier near-global DEMs, such as SRTM, to extend the temporal baseline of the analysis. Emerging space-borne lidar-based DTMs (e.g., [Pracná et al. 2025](#)) could further help correct vegetation-related biases. Furthermore, the increasing availability of repeated airborne lidar acquisitions, particularly in Europe and North America, opens

opportunities for more precise monitoring of mine depth and volume change over time (Moudrý et al. 2026).

In addition, improving mining area detection remains a key challenge. While many studies rely on manual digitization of mines from satellite imagery (e.g., Daham et al. 2024), automated approaches that integrate multiple remote sensing datasets have shown considerable promise (e.g., Yu et al. 2018; Wu et al. 2020; Daichendt et al. 2025). Integrating mine volume information derived from DEMs of difference with high-resolution imagery could further improve the accuracy and timely detection of surface mining operations globally.

Future research should examine how differences in extracted and deposited volumes translate into distinct environmental and social impacts across different jurisdictions. Integrating global mining volume datasets with spatial data on settlements, infrastructure, water systems and biodiversity could help identify emerging “boundary zones” where mining expansion increasingly intersects with communities and ecosystems. Further work is also needed to assess how national governance frameworks, permitting regimes, and consent processes shape the capacity of countries to manage the social and environmental implications of large-scale mining volumes.

5. Conclusions

This global assessment advances our understanding of the complex footprint of surface mining. By incorporating volumetric estimates, rather than relying solely on areal measures, this study introduces a more nuanced and physically meaningful proxy for mining intensity and geomorphological transformation. The results reveal differences across continents, countries, biomes and mining sites that area alone cannot explain.

Beyond its empirical contributions, this work has notable implications for research, policy and practice. First, volume-based metrics can enhance global environmental monitoring by providing more accurate indicators of landscape disturbance and habitat loss, dimensions that are often underestimated using only areal measures. Second, the dataset serves as a robust foundation for more spatially explicit assessments of the social impacts of mining and waste management. Third, the results offer valuable

insights for policymakers and regulators seeking to refine land use planning, mine closure strategies and rehabilitation standards (e.g., in the framework of the UN Convention to Combat Desertification). Cross-national, cross-regional and cross-site comparisons reveal important variations in mining practices that can inform more context-specific and effective policy responses. Finally, by addressing the long-overlooked vertical dimension of mining activity, this study fills a critical data gap and sets a precedent for future efforts to capture the full scale of anthropogenic landscape change through mining.

Data Availability

Data supporting this study are freely available. The Global-scale mining polygons dataset is available from PANGAEA: <https://doi.org/10.1594/PANGAEA.942325> (Maus et al. 2022; version 2). The TanDEM-X 30m DEM Change Maps are publicly available at <https://geoservice.dlr.de/web/maps/tdm:dc30> (Lachaise et al. 2022). Data produced within this study are available in a public repository and will be made openly accessible upon publication. During review, they can be accessed via the following anonymous link: [\[LINK\]](#).

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Tables

Table 1. Global mining volume (extracted plus deposited volume) per biome and continent. Values represent the ratio between a biome's share of total mining volume and its share of total land area within each continent. Values > 1 indicate disproportionately high mining intensity relative to area, whereas values < 1 indicate lower intensity.

Continent / Biome	Boreal Forests /Taiga	Deserts & Xeric Shrub.	Flooded Grasslands & Savannas	Mangroves	Mediterranean Forests, Woodlands & Scrub	Montane Grasslands & Shrublands	Temperate Broadleaf & Mixed Forests	Temperate Conifer Forests	Temperate Grasslands, Savannas & Shrublands	Tropical & Subtropical Coniferous Forests	Tropical & Subtropical Dry Broadleaf Forests	Tropical & Subtropical Grasslands, Savannas & Shrublands	Tropical & Subtropical Moist Broadleaf Forests	Tundra
Africa		0.6	0.2	4.4	1.1	6.1		0.3			0.0	0.8	2.1	
Asia	0.4	0.4	0.4	2.3	2.2	1.1	1.3	1.2	1.3	0.1	1.6	0.0	2.8	0.1

Australia		0.5			1.0	0.0	2.9		0.0			1.6	0.1	
Europe	0.3			0.0	0.8		1.3	0.2	1.6				2.3	0.1
North America	0.9	2.2	0.0	0.0	0.1		1.5	1.6	1.0	0.94	1.0	0.0	0.4	0.1
Oceania				0.0		0.8	1.3		1.8		0.2	0.0	0.9	
South America		5.7	0.1	0.2	1.9	2.2	0.2		0.0		0.6	0.2	1.4	

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Declaration of interests

☒The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

☐The authors declare the following financial interests/personal relationships which may be considered as potential competing interests:

Dear Editors,

Please find attached our manuscript entitled “Quantifying Global Material Displacement from Surface Mining” for consideration as a Research Paper in Resources, Conservation and Recycling.

In this study, we provide the first global estimates of extracted and deposited material volumes associated with surface mining. By combining a global mining polygon dataset with TanDEM-X 30 m DEM Change Maps, we quantify terrain changes between 2011–2015 and 2016–2022 and derive spatially explicit estimates of material displacement across continents, countries, and biomes.

Our results show that 279 billion m³ of material were extracted globally and 236 billion m³ were deposited, revealing pronounced geographic asymmetries and highlighting regions of particularly high mining intensity. Importantly, we demonstrate that volume-based indicators provide a complementary perspective to conventional area-based assessments of mining extent, enabling more consistent cross-site comparisons and offering new insights into the three-dimensional footprint of resource extraction.

This manuscript is original, has not been published previously, and is not under consideration for publication elsewhere. Data supporting this study are publicly available and will be fully accessible upon publication.

Thank you for considering our manuscript. We look forward to your response.

Sincerely,

Vítězslav Moudrý, on behalf the co-authors

First global estimates of extracted/deposited material volumes from surface mining

279 billion m³ of material extracted, dominated by Asia (133) and South America (62)

Highest extraction volumes occur in Indonesia, China, Russia, Australia, and Brazil

In several biomes, extraction volumes exceed expectations based on geographic extent

Mining volumes provide a complementary measure of mining intensity to site area

Supporting Information for

Quantifying Global Material Displacement from Surface Mining

Contents of this file: Figures S1 – S3; Tables S1 – S3

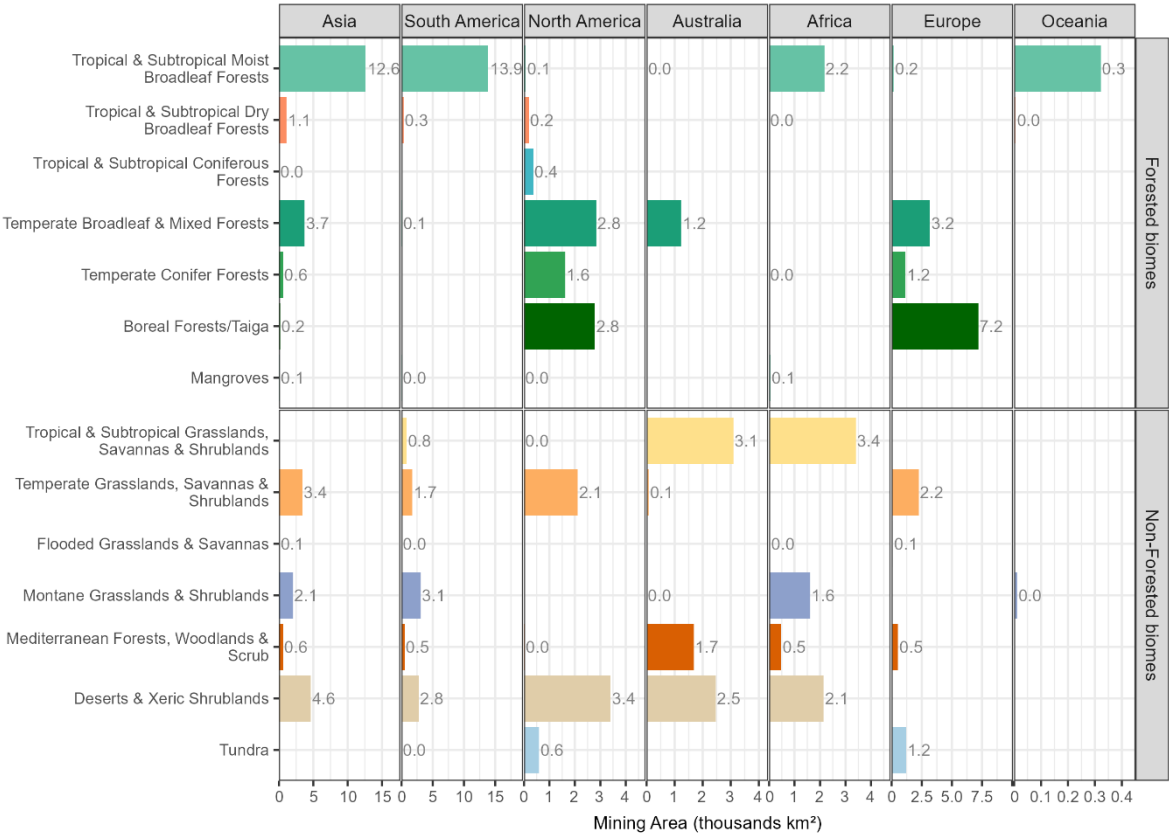


Figure S1. Bar chart showing the total area of mining polygons across seven forested and seven non-forested biomes on each continent. Note the different scales on the x-axis for each continent.

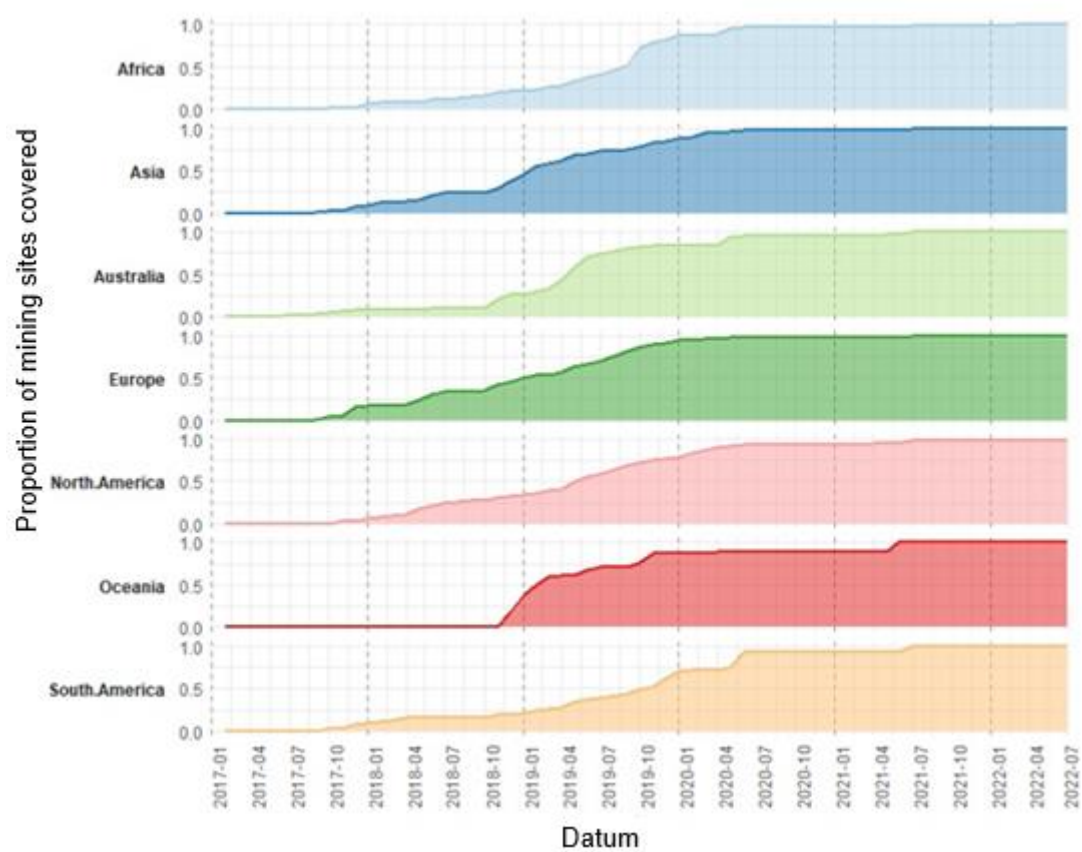


Figure S2. Cumulative development of mining sites coverage by TanDEM-X Change Maps between 2016 and 2022 for individual continents.

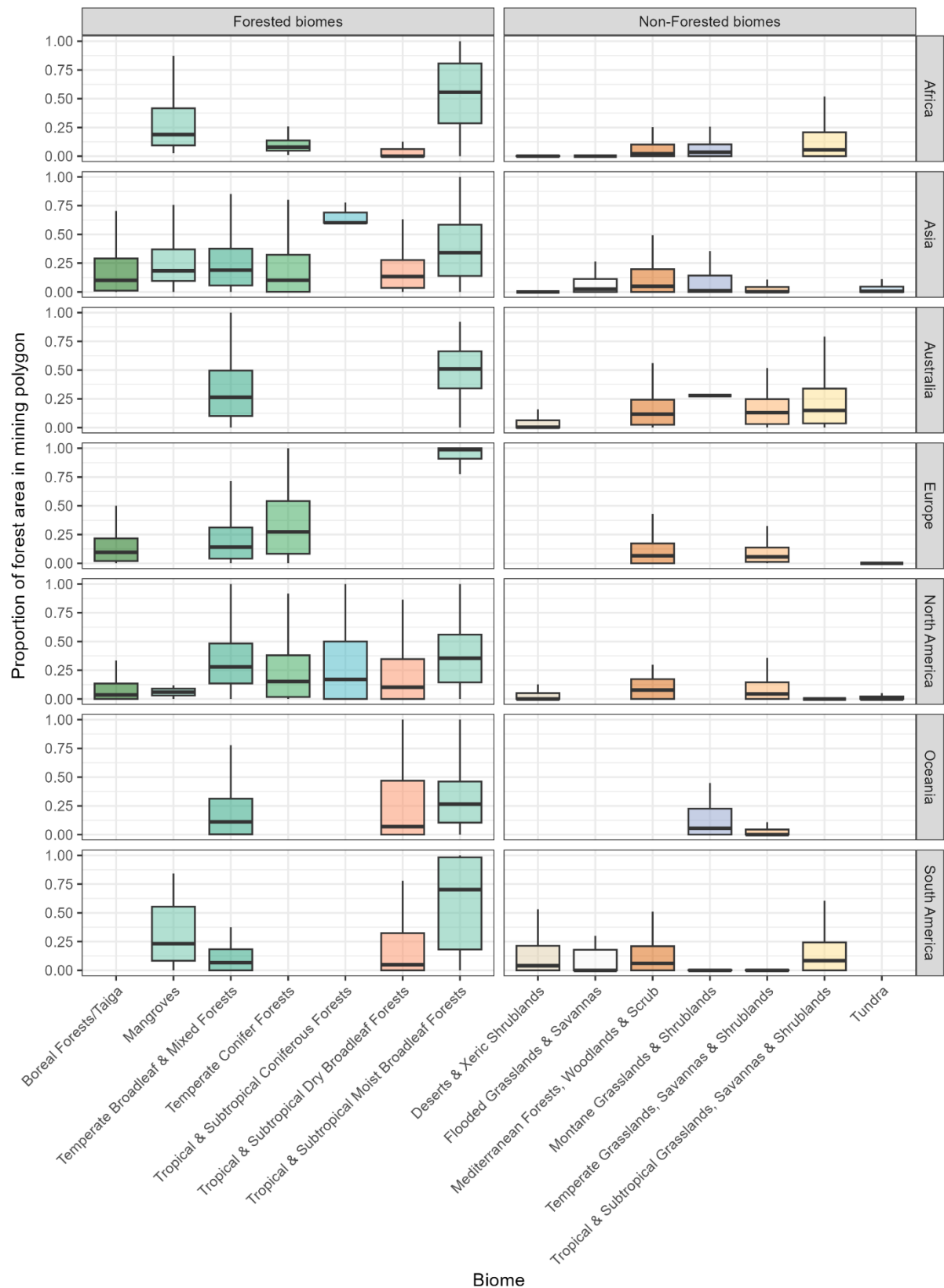


Figure S3. Box plots showing the relationship between the proportion of forest cover within mining polygons (derived from land cover mask; LCM), biome, and continent.

Table S1. Mining areas with zero deposited volume and non-zero extracted volume

Continent	Count	Area (km sq)	% count (continent)	% area (continent)
Africa	338	26.04	6.84	0.26
Asia	720	49.10	3.94	0.12
Australia	180	22.85	5.27	0.27
Europe	142	4.08	3.92	0.10
North America	604	34.05	7.05	0.24
Oceania	23	0.66	5.78	0.15
South America	263	13.57	4.62	0.06
Total	2270	150.36		

Table S2. Mining areas with zero extracted volume and non-zero deposited volume

Continent	Count	Area (km sq)	% count (continent)	% area (continent)
Africa	307	33.14	6.21	0.33
Asia	655	46.40	3.58	0.11
Australia	230	26.31	6.73	0.31
Europe	143	6.22	3.95	0.15
North America	418	24.32	4.88	0.17
Oceania	14	0.41	3.52	0.09
South America	260	17.91	4.57	0.08
Total	2027	154.72		

Table S3. Mining areas with no change (both deposited and extracted volumes were zero)

Continent	Count	Area (km sq)	% count (continent)	% area (continent)
Africa	296	19.41	5.99	0.20
Asia	277	14.92	1.51	0.04
Australia	244	15.14	7.14	0.18
Europe	38	0.57	1.05	0.01
North America	389	14.29	4.54	0.10
Oceania	5	0.09	1.26	0.02
South America	244	10.36	4.29	0.04
Total	1493	74.79		

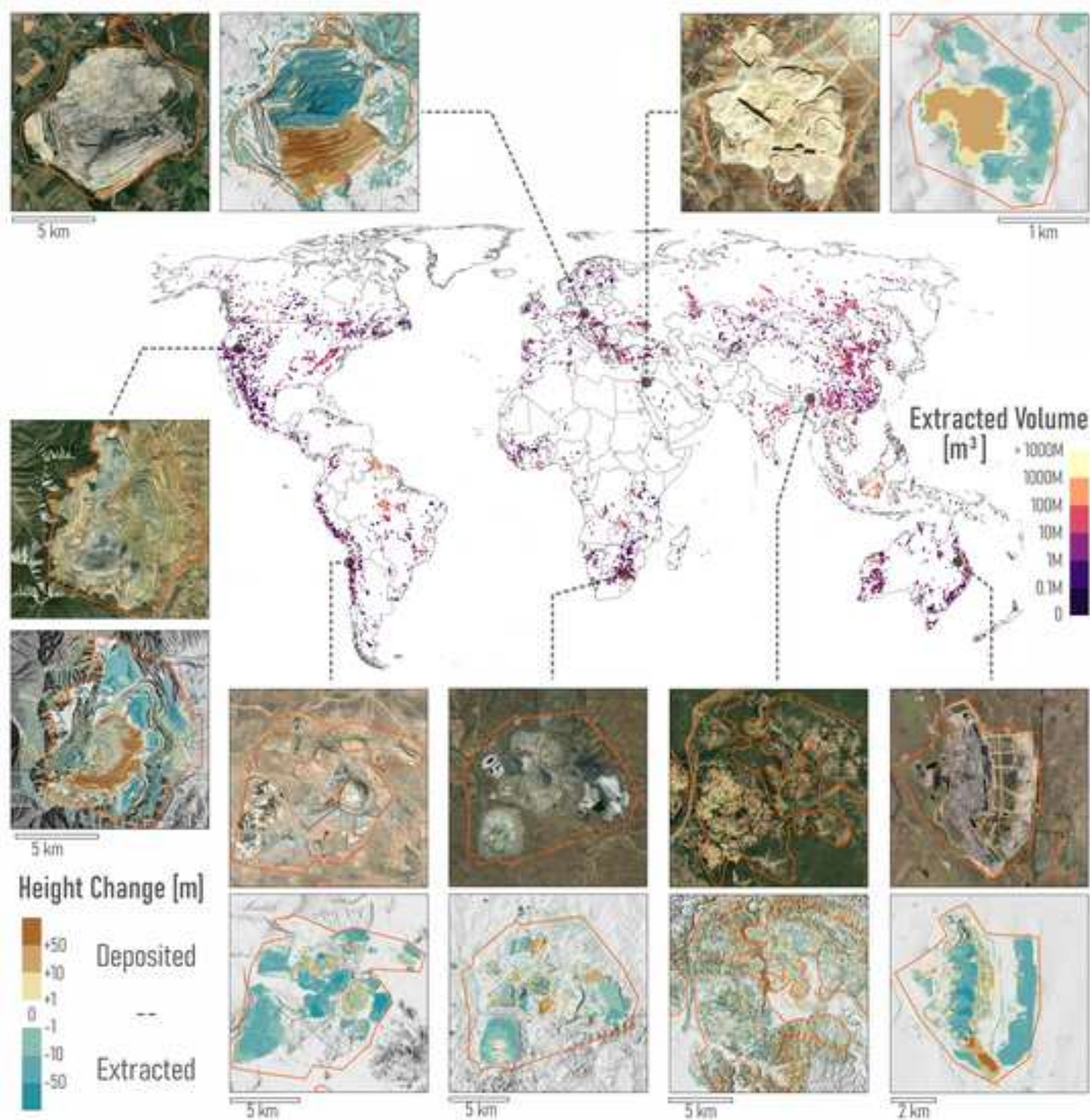


Figure 1

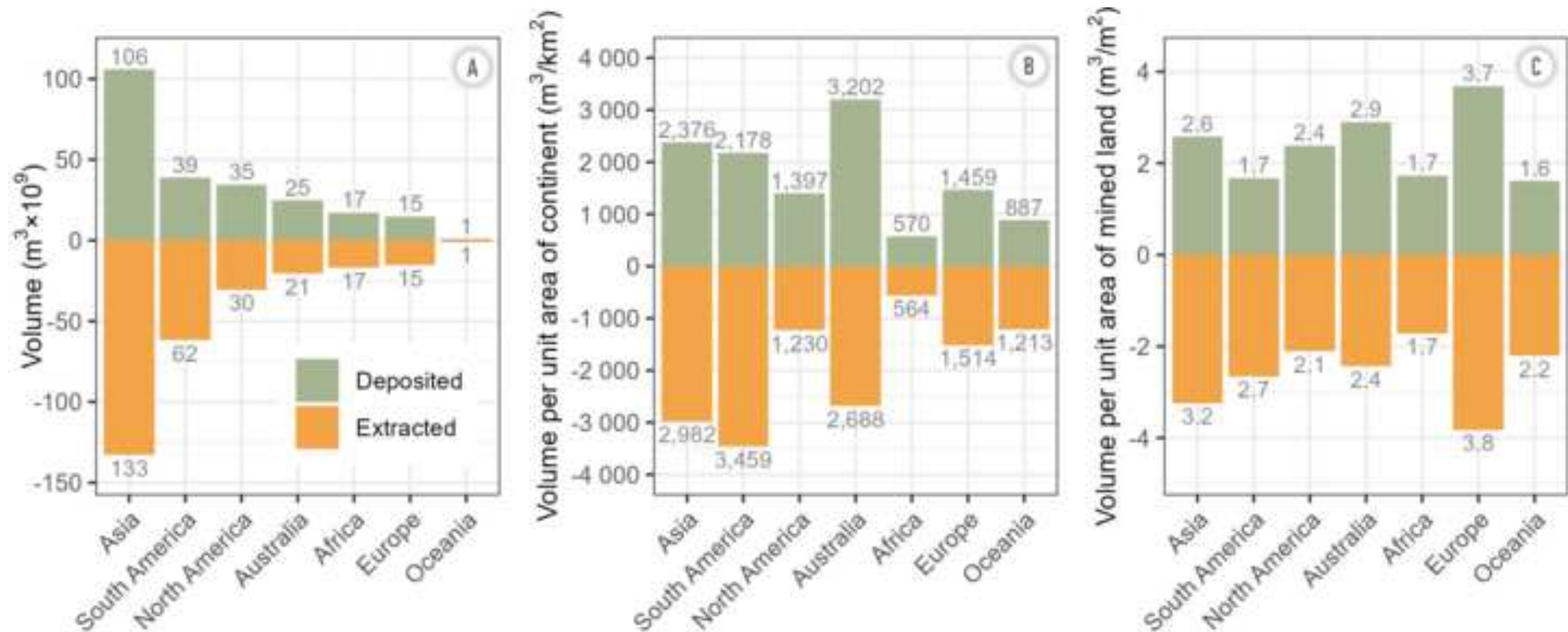


Figure 2

[Click here to access/download;Figure\(s\);Figure_2.jpg](#)

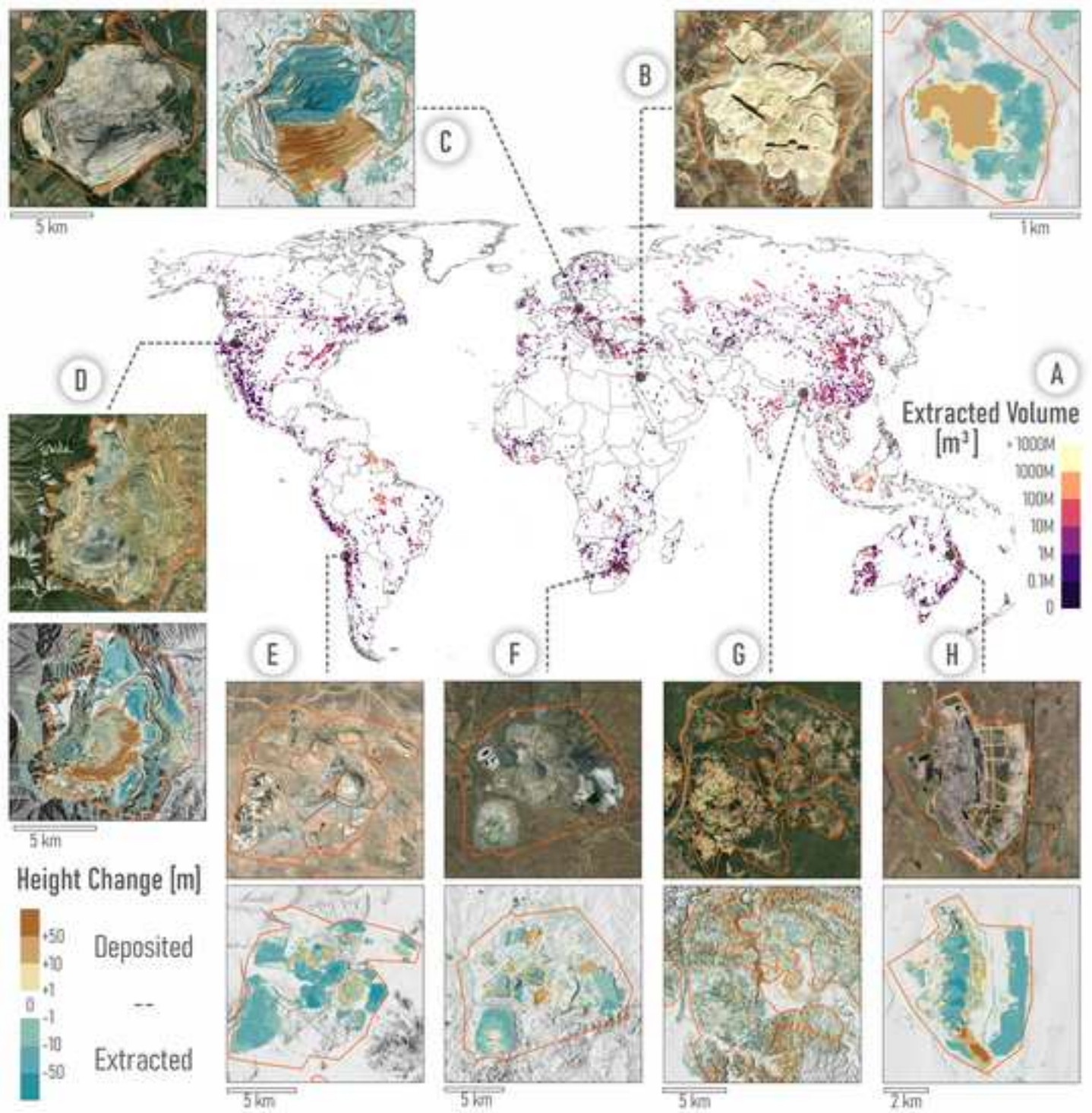


Figure 3

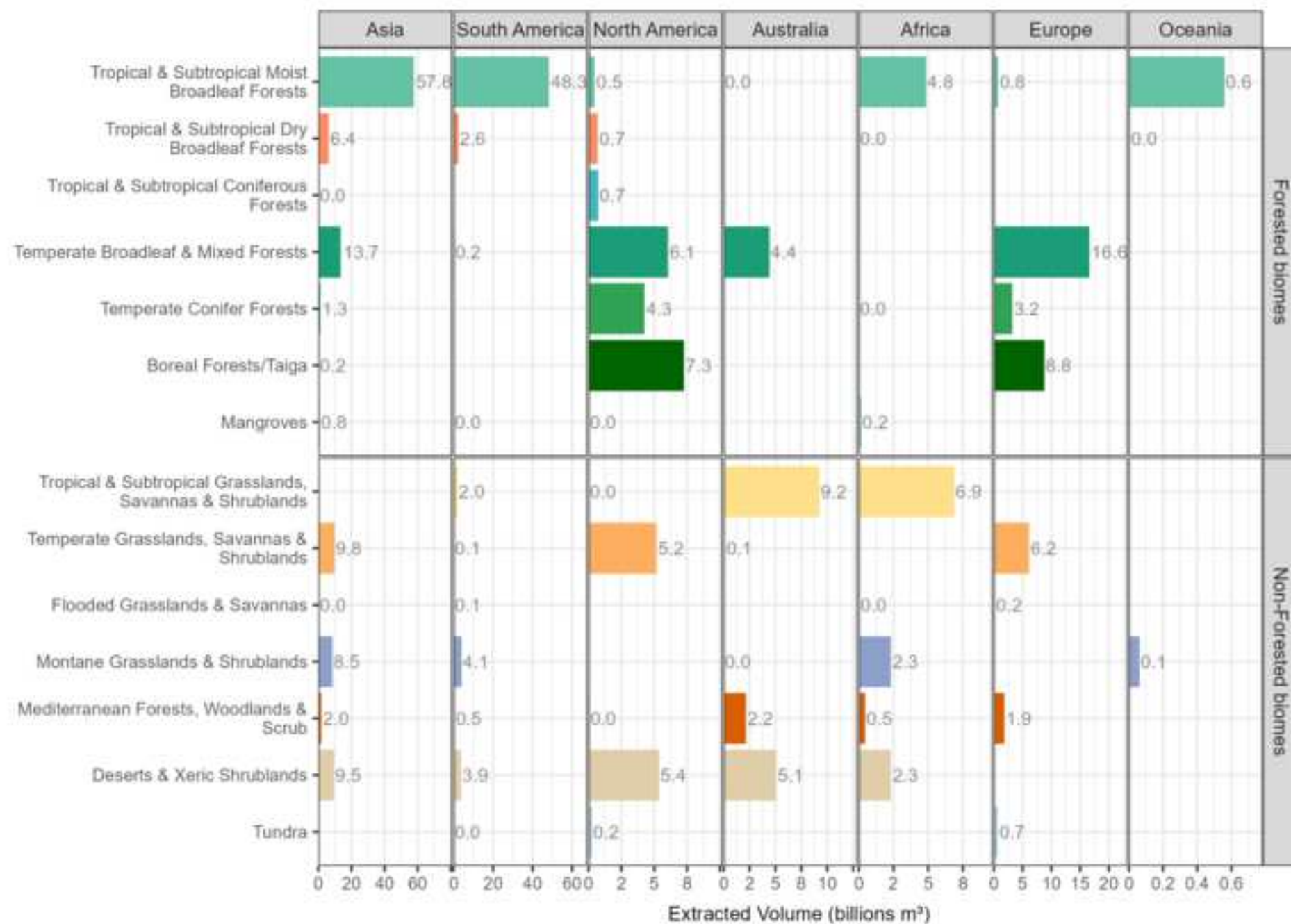
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Figure 4

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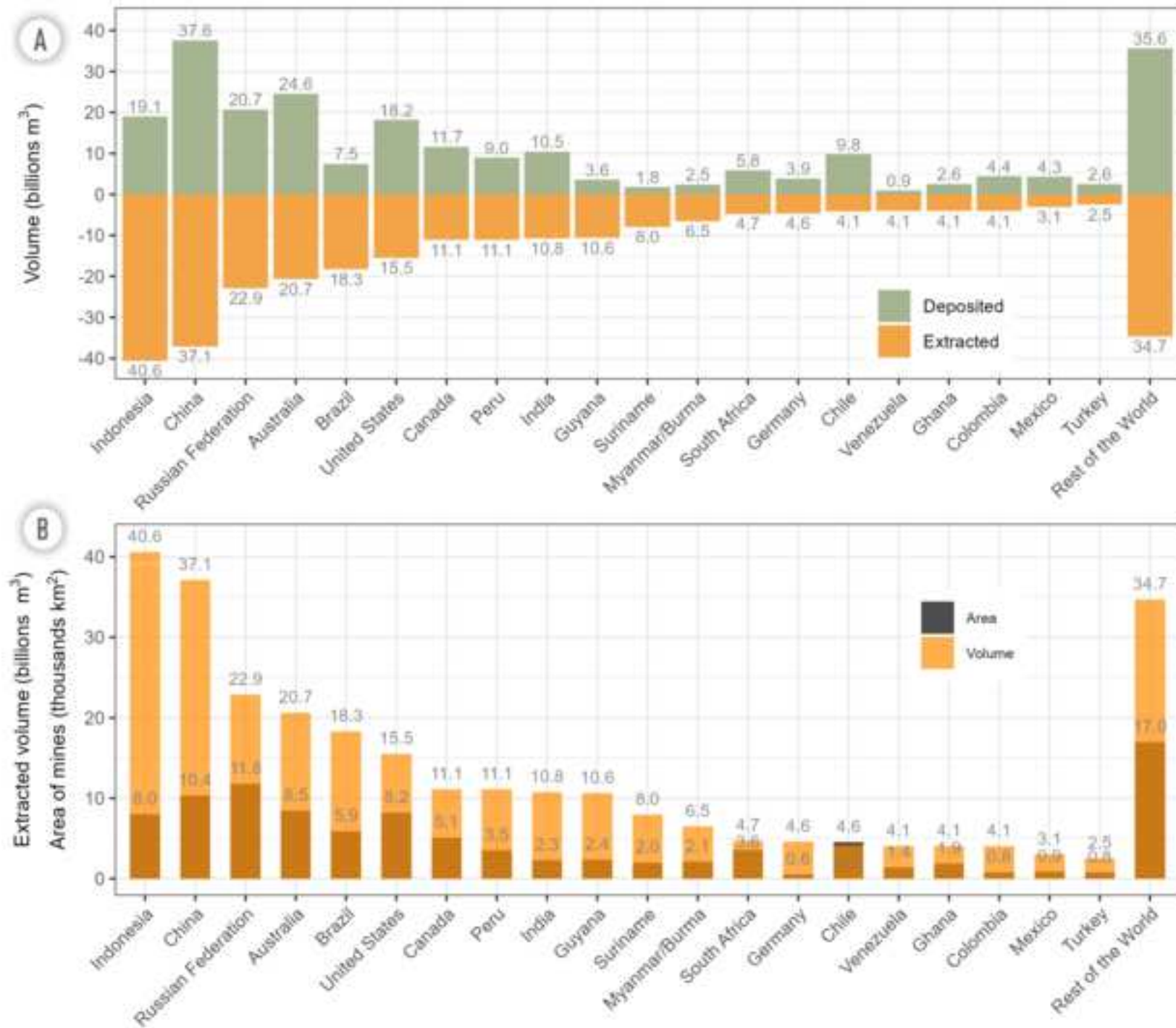


Figure 5

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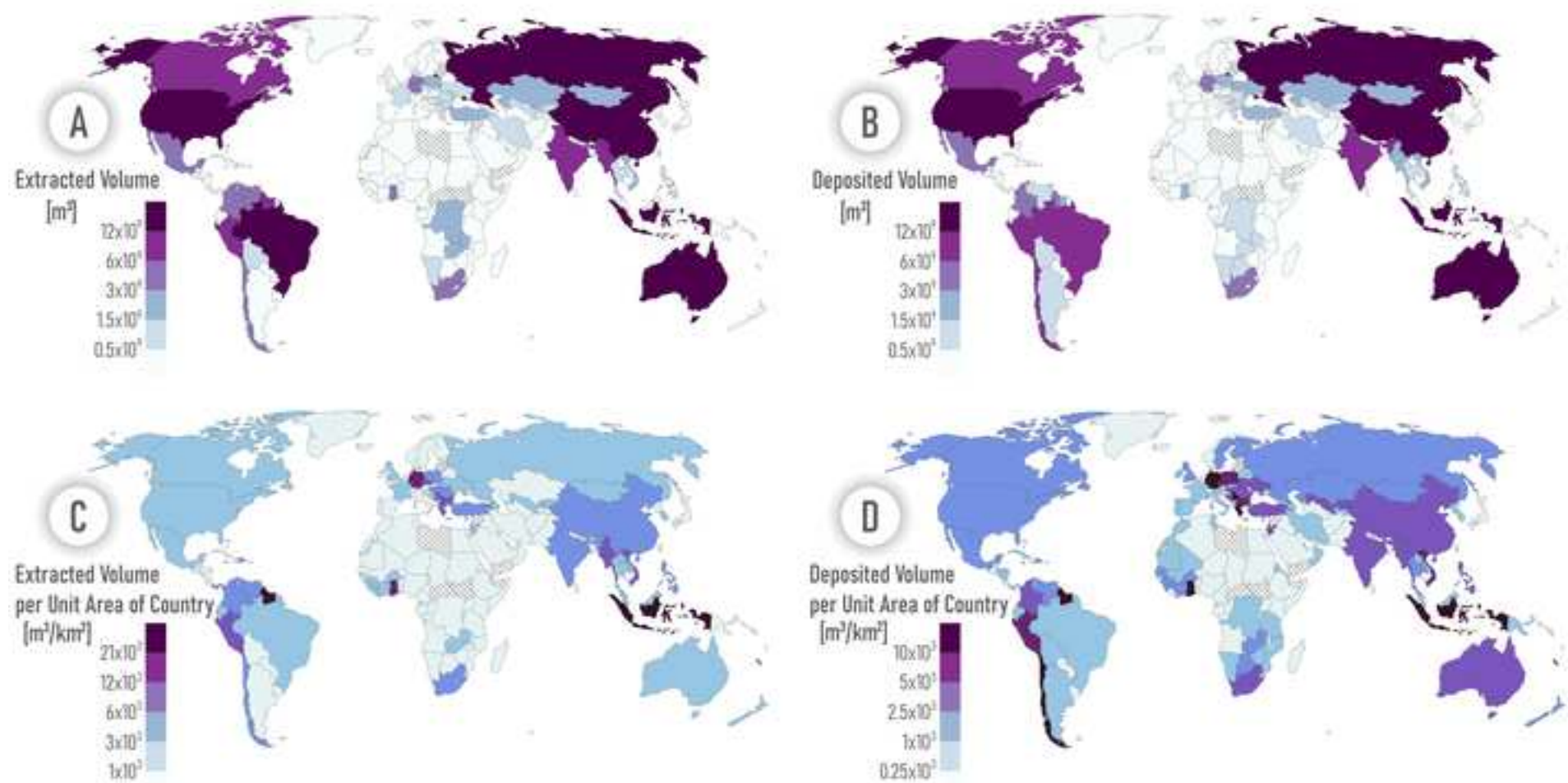


Figure 6

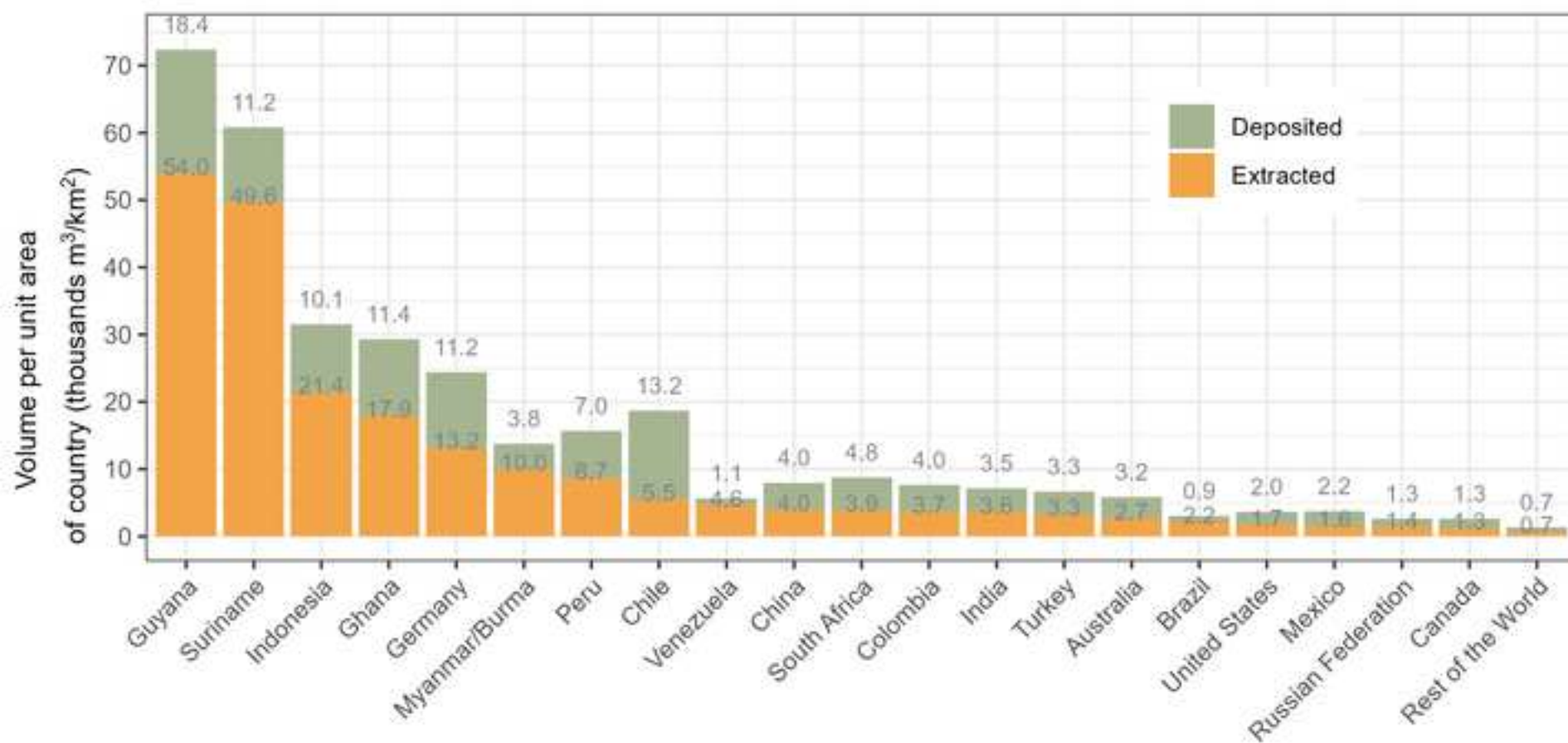


Figure 7

